Pablo Alejandro López Sánchez

Data Engineer · AI & Automation

Manizales, Colombia | +57 318 766 2292 | poglolopez@gmail.com | Portfolio | LinkedIn | GitHub

PROFILE

Data Engineer specialized in automation and AI systems. I design and implement data architectures, MLOps infrastructure, and automation pipelines, and bring AI to the core of operations through agentic systems and models that turn manual processes into autonomous workflows. I build technical projects end-to-end, from data extraction through modeling, orchestration, and interface, integrating LLMs to automate analysis, generate insights, and support business decisions at scale.

SKILLS

Data Engineering: Python (Polars, SQLAlchemy, FastAPI, sqlglot), SQL (SQL Server, PostgreSQL), ETL Pipelines, Web Scraping (Selenium, Scrapy).
AI & Agentic Systems: Multi-agent development, spec-driven development, harness engineering, MCP (Model Context Protocol, FastMCP), OAuth 2.0, Claude Code / Agent SDK, prompt engineering.
Infrastructure & Orchestration: Dagster, Docker, Cloudflare Tunnel, SSH/SFTP, Linux Server, Git/GitHub, Data Flow Modeling.
Analytics: Power BI (DAX, Power Query), KPI Analysis.
Business Strategy: Project Management, Demand Planning, Inventory Optimization.
Languages: English (B2-C1), Spanish (Native).

EXPERIENCE

Data Engineer – Mercaldas, Manizales, Colombia
July 2025 – Present

Leading the company's data and automation engineering, driving end-to-end the shift from siloed, manual processes to a centralized architecture with in-house MLOps, autonomous demand forecasting, price intelligence, and an MCP server connecting Claude to the data warehouse so users can query and export data without writing SQL, plus internal tools already adopted across multiple business areas, including senior leadership.

- **MLOps Infrastructure & Data Autonomy:** Led the design and implementation of an end-to-end MLOps infrastructure for autonomous demand forecasting, reaching **70-82%** forecast accuracy depending on the cluster and driving a decisive reduction in **stockouts** from a historical **12-20%** range to around **8%**.
- **MCP Server for the Data Warehouse (Claude):** Designed and deployed an MCP server that connects Claude to the data warehouse, letting C-levels and team leaders query data in natural language, generating their own reports. It improved analytics and information availability across the company.
- **Market Price Intelligence:** Designed and built an internal analytics platform that forecasts wholesale agricultural product prices **52 weeks** ahead using per-cluster ML models (**XGBoost/LightGBM**) and generates **automated AI insights** (Claude), delivered through a custom web app that benchmarks the market against internal prices to anticipate purchasing decisions.
- **Stock-Transfer Planning Automation:** Built the web tool the **purchasing analysts** use to plan inventory transfers between stores; it cuts each transfer plan from hours to minutes (**1-4 hours saved per run**) and improves merchandise-reallocation decisions, reducing overstock and stockouts across stores.

- **Centralized Orchestration (Dagster):** Deployed and configured an orchestration platform on a dedicated **Linux** server to unify dispersed processes. Designed to automate manual executions and optimize failure monitoring, resulting in an estimated reduction of ~4 operational hours per week per user for the IT and Demand Planning teams.

Business Analyst – Latin America Sourcing, Remote
May 2025 – July 2025

- Managed and analyzed commercial data using Airtable and Stripe to optimize negotiation processes with international suppliers.
- Analyzed client requests, documented commercial interactions, and generated operational insights. Automated supplier classification processes using custom Python scripts.

Independent Financial Analyst – Independent, Manizales, Colombia
Jan 2024 – Dec 2024

- Technical and market analysis to trade with my own capital.
- Tracked my own performance in Excel (trade metrics and correlation analysis), with systematic logging of every trade and periodic reviews to adjust my rules.
- Risk management: stop losses, daily loss and drawdown limits, and portfolio diversification.

Business Intelligence Internship – Industria Licorera de Caldas, Manizales, Colombia
Feb 2023 – Jul 2023

- Developed Power BI dashboards for Operations and Foreign Trade departments, integrating data from Dynamics 365 to drive logistics efficiency.

PROJECTS

Five production systems, each with an interactive walkthrough: pablo.maieutik-data.com.

ML demand forecasting: Weekly forecasts for 100,000+ product and store combinations across 14 stores, feeding replenishment. 19% lower error than the system it replaced.
MCP server for the data warehouse: Connects Claude to the data warehouse so C-levels and area leaders can query and export data in plain language, with corporate login and an audit trail of every query.
Market price intelligence: Ingests Colombia's weekly wholesale price bulletins, forecasts 52 weeks per product, and has a model write the reading of each series.
Internal operations platform: Central API for recurring operations work. Transfer planning went from roughly 4 hours to 2-5 minutes, run by the team itself.
Personal AI operating system: Personal project. A Git-versioned Markdown vault as long-term memory, with a multi-agent runtime that asks before anything it cannot undo.

EDUCATION

Bachelor of Business Administration (BBA) – Universidad Nacional de Colombia, October 2023

CERTIFICATES & COURSES

Full certificates list: here

Google Data Analytics Professional Certificate – Google | Coursera
Databases for Data Scientists Specialization – University of Colorado Boulder | Coursera
Claude Code in Action – Anthropic Education
Data Analysis Tools – Wesleyan University | Coursera

REFERENCES

Luisa Soraya Santana Murcia – Mercaldas | Demand Planning Leader, +57 318 358 9892, ssantana@mercaldas.com.co
Nicolas Navarro Felizzola – itti | Data Engineer, +57 317 557 6781, iniconavarro@gmail.com
Juan Luis López Toro – Mercaldas | Supply Manager, jlopez@mercaldas.com.co